Estimating heterogeneous treatment effects using machine learning

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Motivation

There are a number of reasons that we might care about subgroup treatment effects for an experiment

- Subgroup affects allow for better, more precise causal inference.
- To better target some policy intervention.
- To understand what mechanisms are at work, though this might require some additional assumption (Fu and Slough 2026).

From ATEs to CATEs

Each observation i has two **potential outcomes:** $Y_i(1)$ if treated, $Y_i(0)$ if not. We observe only one:

$$Y_i = W_i \, Y_i(1) + (1 - W_i) \, Y_i(0)$$

so the individual effect $Y_i(1) - Y_i(0)$ is never observed.

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$$\tau = \mathbb{E}\big[Y_i(1) - Y_i(0)\big] = \mathbb{E}[Y_i \mid W_i = 1] - \mathbb{E}[Y_i \mid W_i = 0]$$

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But could we estimate conditional average treatment effects (CATEs)?

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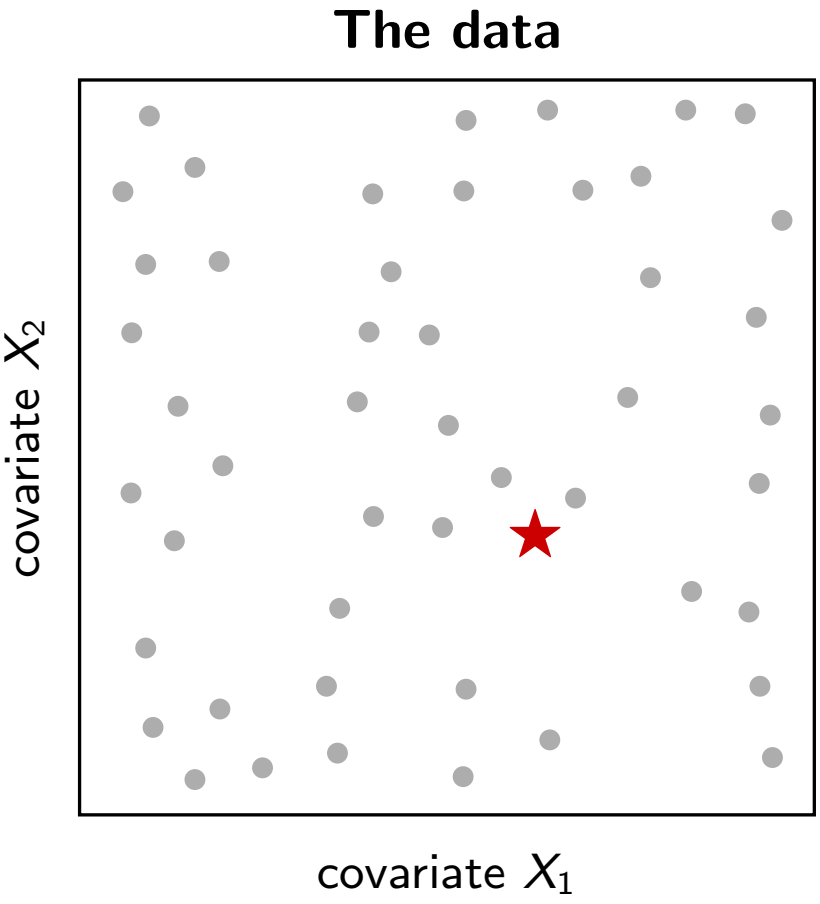
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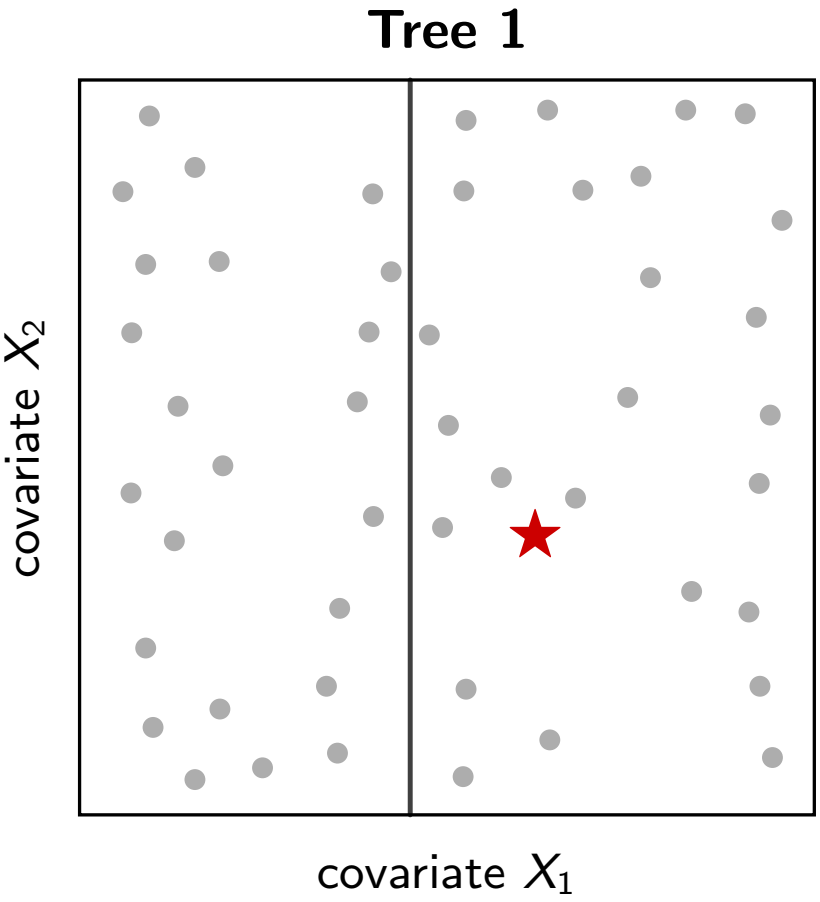
With many covariates, the **curse of dimensionality** leaves almost no observation near any given x. We must borrow from similar units, but similar on which covariates?

Instead of a regression, use a random forest



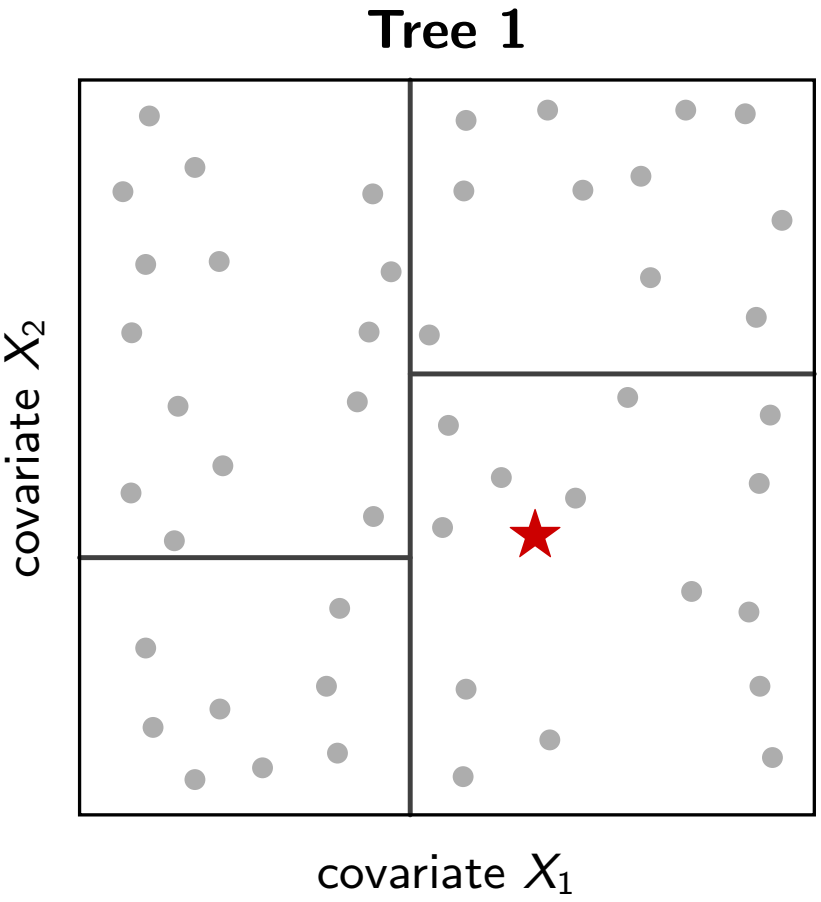
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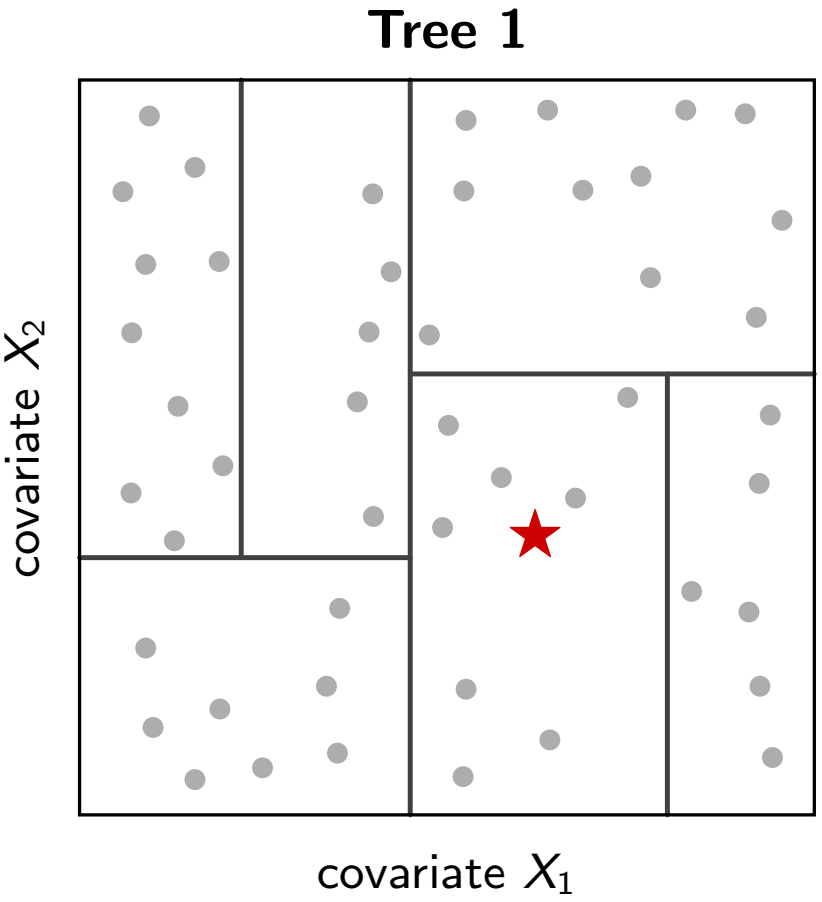
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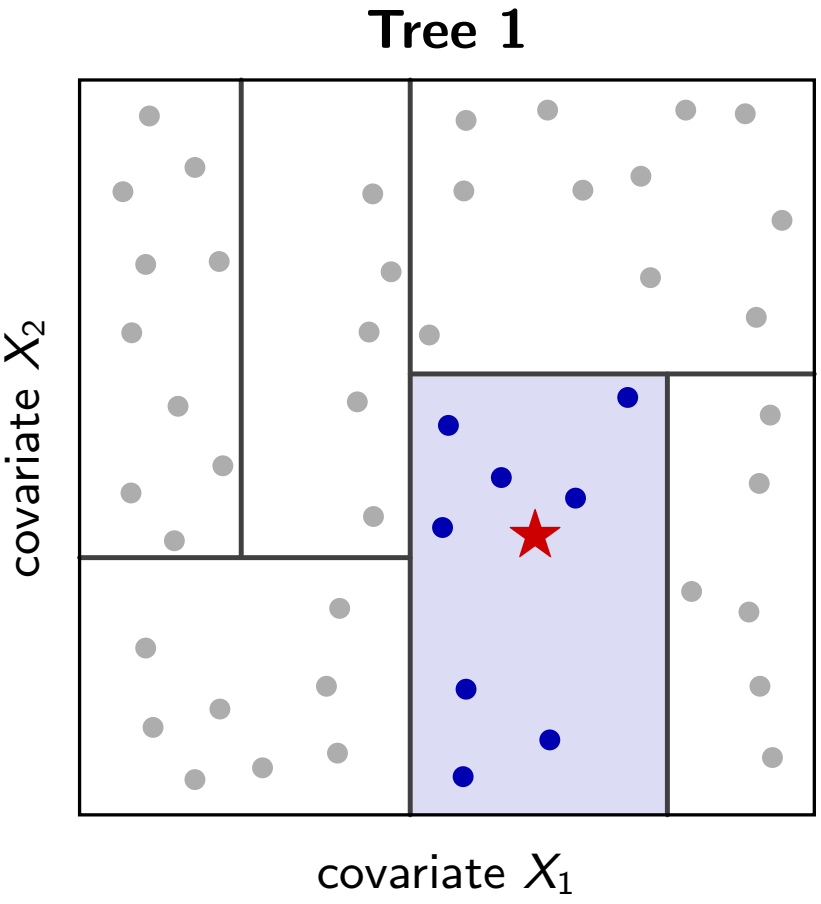
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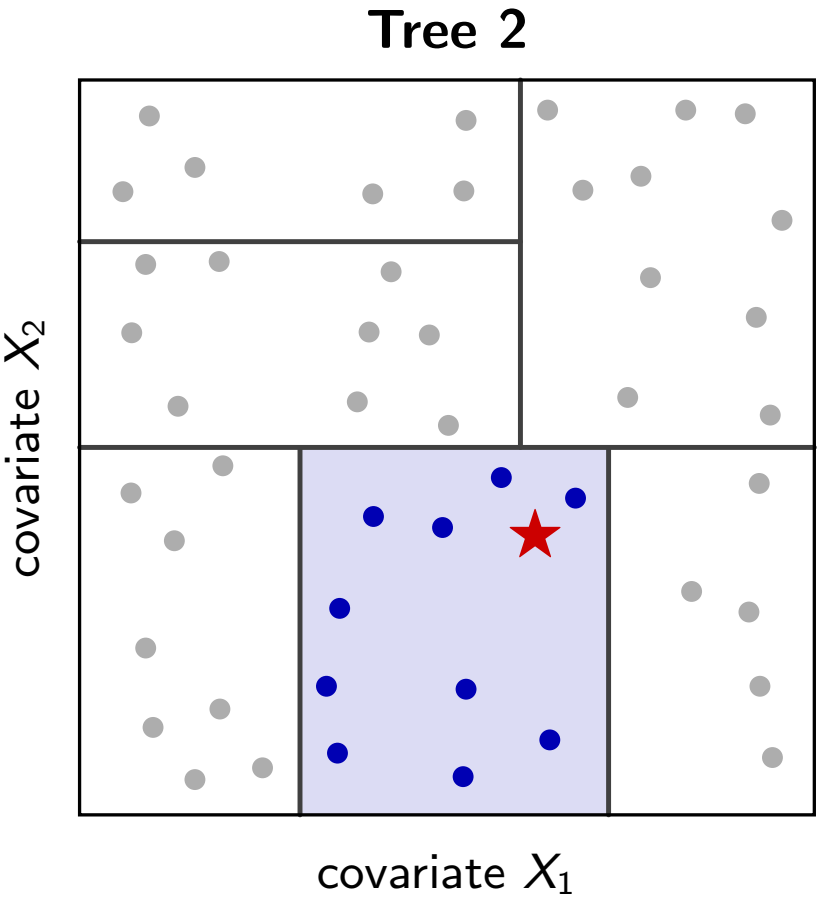
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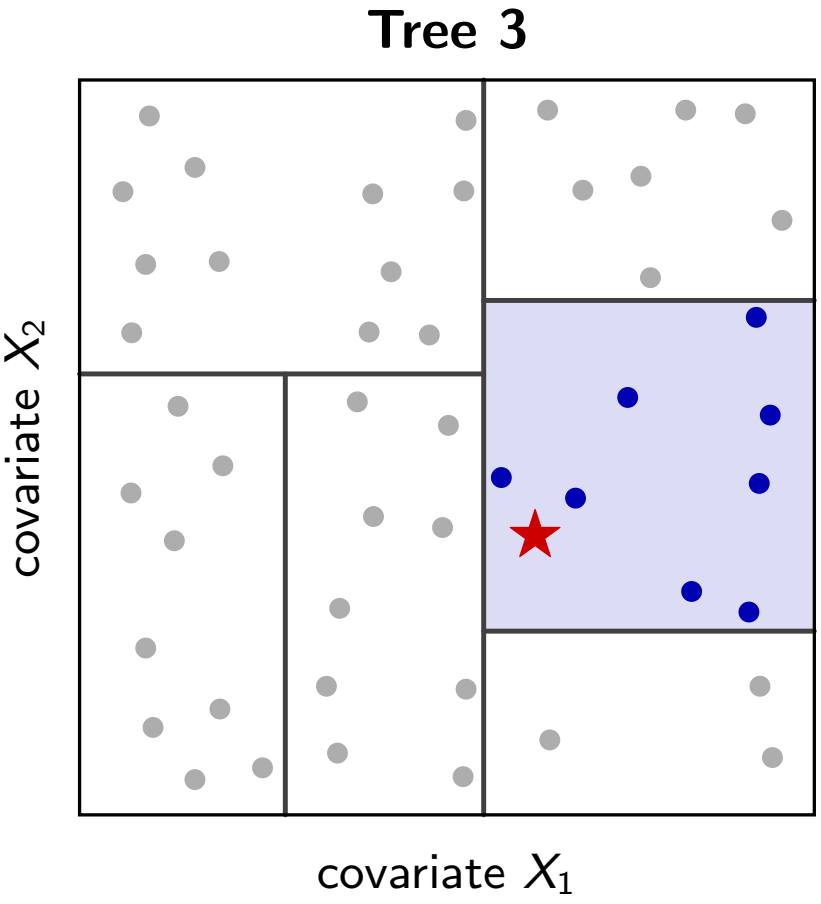
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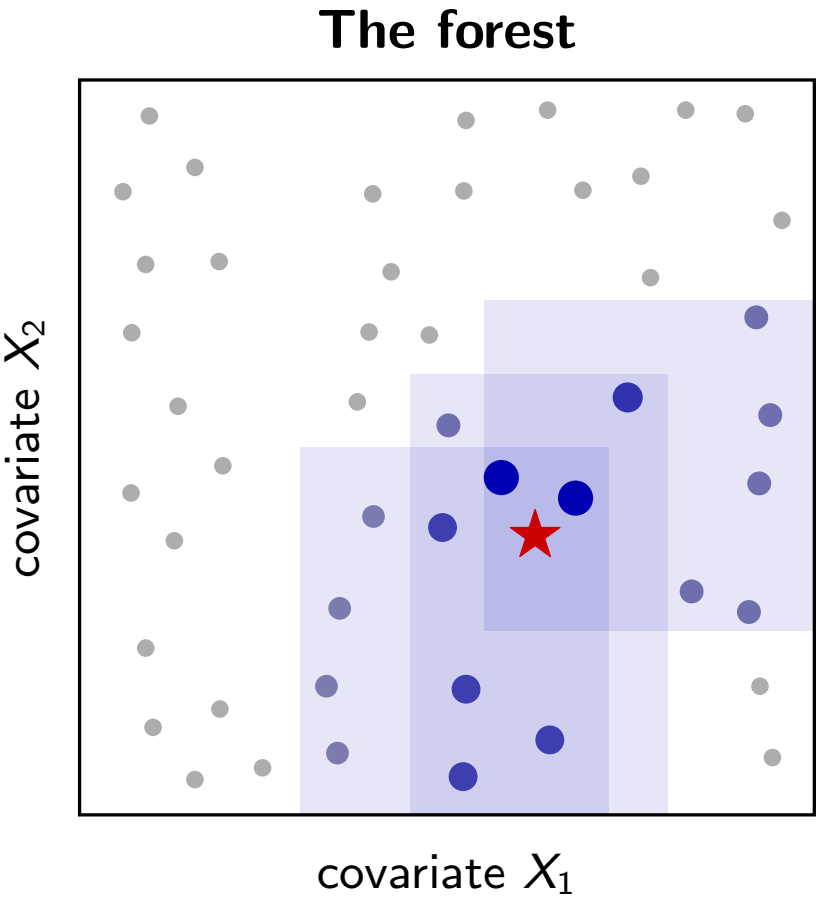
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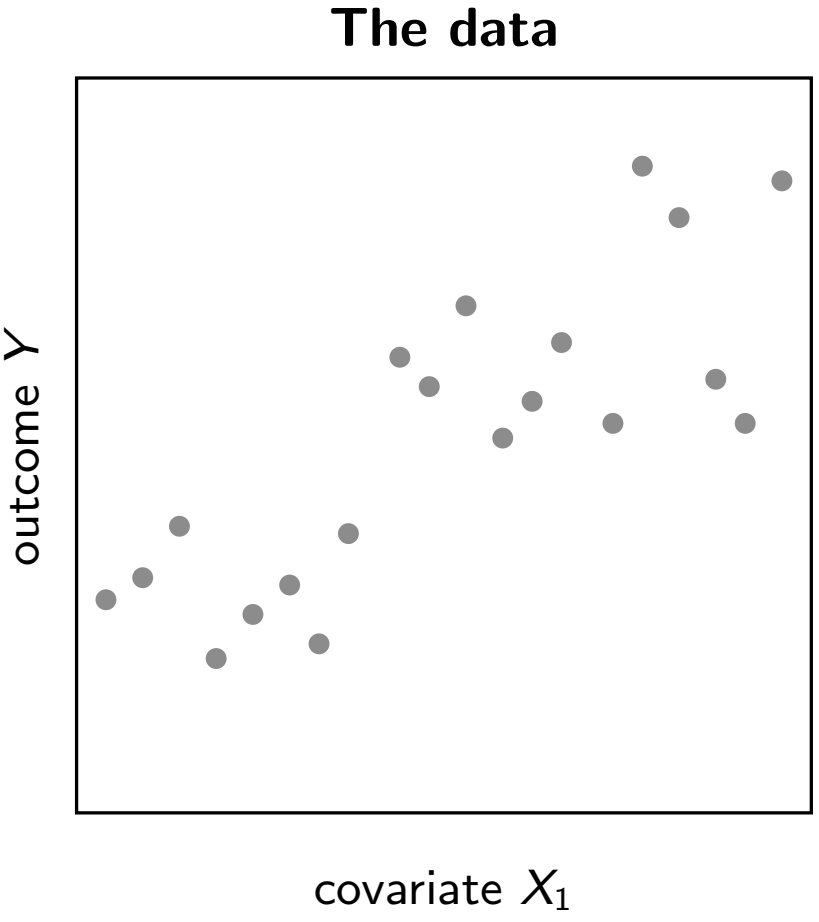
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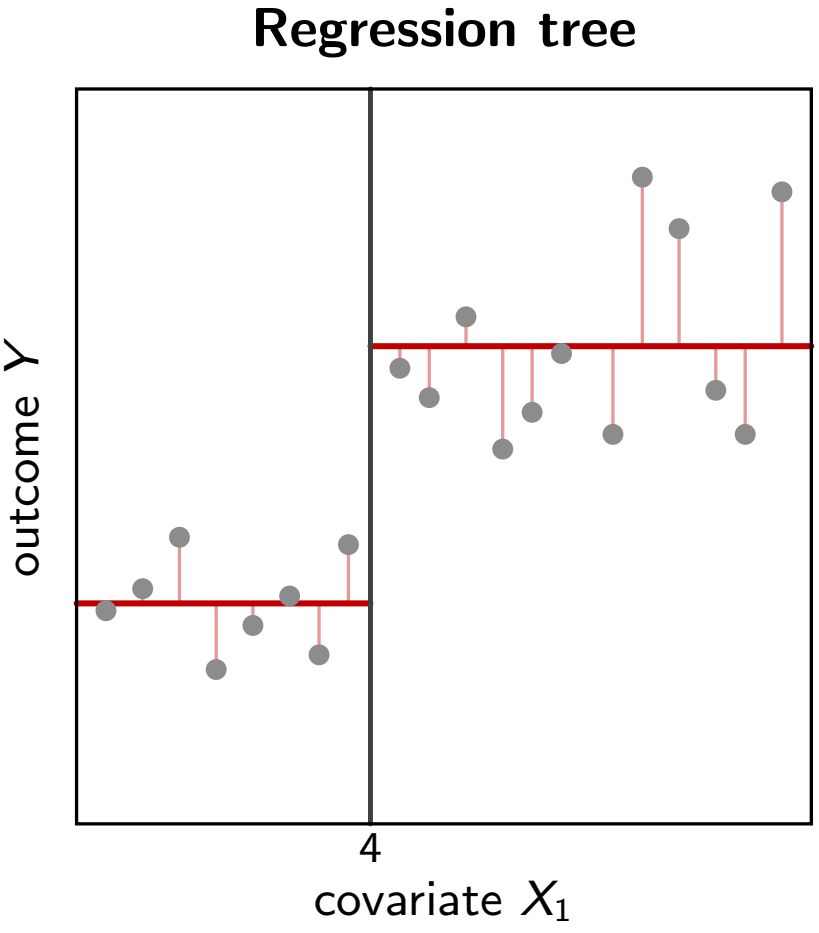
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- Each tree sees a different random subsample, so it cuts differently.
- The more often an observation is a neighbor of x, the more its outcome counts toward the prediction at x.

How does a tree decide where to cut?



- A tree tries every possible cut and keeps the best one. Best at what?

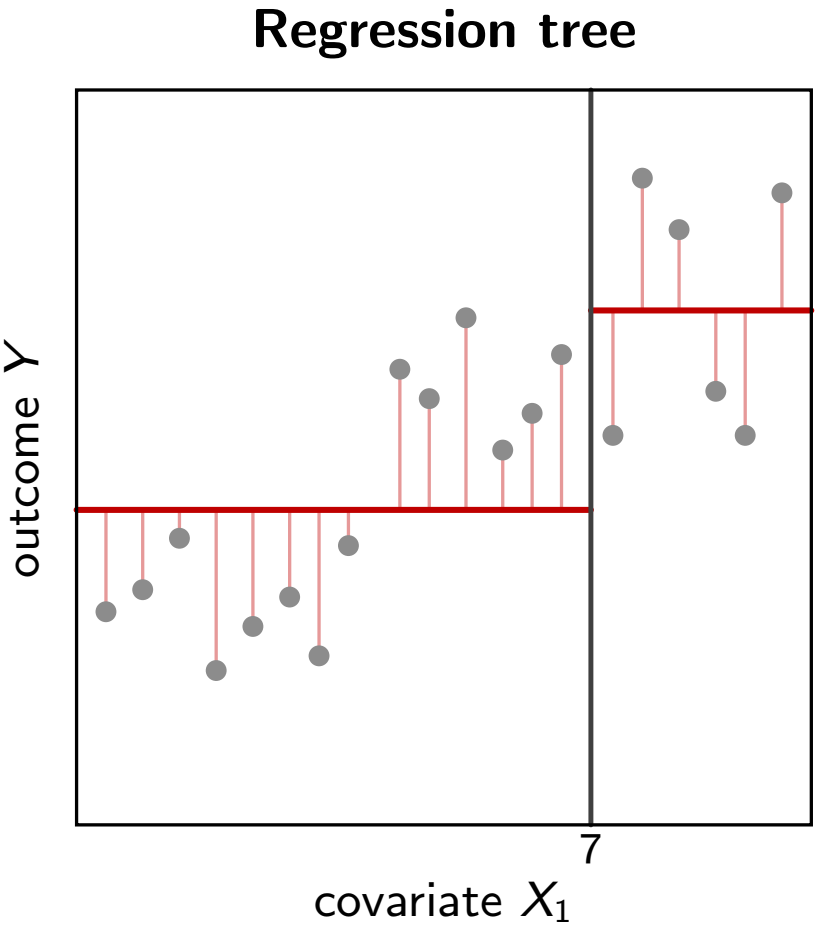
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Variance within nodes: 1.1

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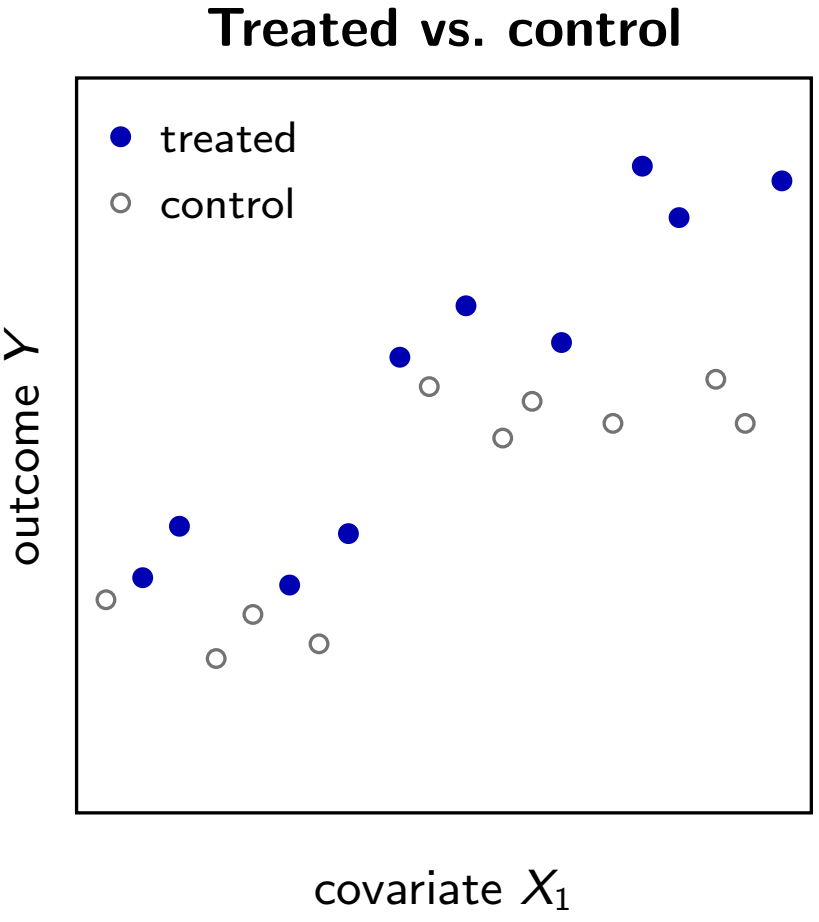
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Variance within nodes: 2.5

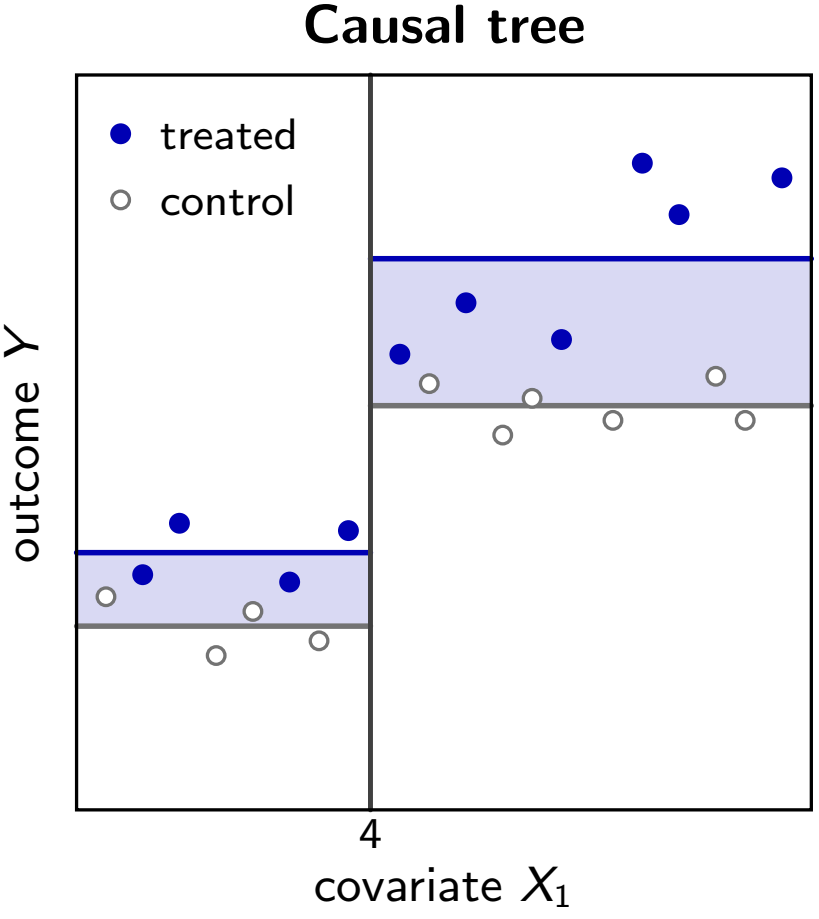
- A tree tries every possible cut and keeps the best one. Best at what?
- **Regression tree:** least variance in Y within each node. Here: cut at 4.

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- **Causal tree:** least variance in treatment effects within each node.

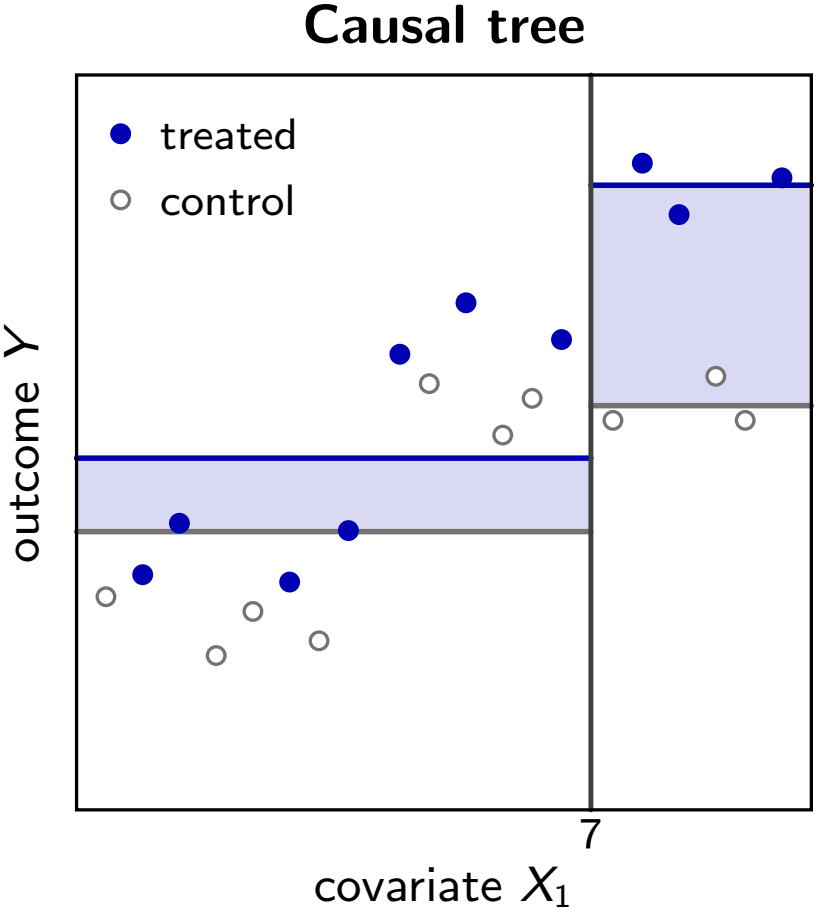
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Effects: 1.0 (left), 2.0 (right)

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How does a tree decide where to cut?



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- A tree tries every possible cut and keeps the best one. Best at what?
- **Regression tree:** least variance in Y within each node. Here: cut at 4.
- **Causal tree:** least variance in treatment effects within each node. Here: cut at 7.

Each tree is grown on a random subsample of the data, and at each split it considers a random subset of the covariates.

Causal Forests: Growing the Trees

Goal: estimate $\tau(x)$, the average treatment effect for units with covariates x.

1. **Center the data.** Replace Y_i and W_i with their residuals after predicting each from X_i (not necessary if you know W_i).
2. **Grow many trees.** For each tree:
   - Draw a random subsample and split it into halves $\mathcal{J}_1$ and $\mathcal{J}_2$.
   - Grow the tree on $\mathcal{J}_1$. At each node:
     - Estimate the node's treatment effect $\hat{\tau}_P$.
     - Give each unit a pseudo-outcome ρ_i: does it pull $\hat{\tau}_P$ up or down? What is ρ_i
     - Split where the children differ most in average ρ_i.
   - Reserve $\mathcal{J}_2$ for estimation ("honesty").

Causal Forests: From Trees to Estimates

3. **Compute weights for a target point x.**
   - In each tree, find the $\mathcal{J}_2$ units that share a leaf with x.
   - Average over trees: $\alpha_i(x)$ measures how often unit i is a neighbor of x.

4. **Calculate one forest, but many regressions.** Weighted regression of centered Y on centered W:

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x)\,(W_i - \bar{W}_\alpha)(Y_i - \bar{Y}_\alpha)}{\sum_i \alpha_i(x)\,(W_i - \bar{W}_\alpha)^2}$$

5. **Quantify uncertainty.** $\hat{\tau}(x)$ is approximately normal, with variance estimated by comparing groups of trees. Variance

Estimating the Nuisance Parameters

```
require(grf)

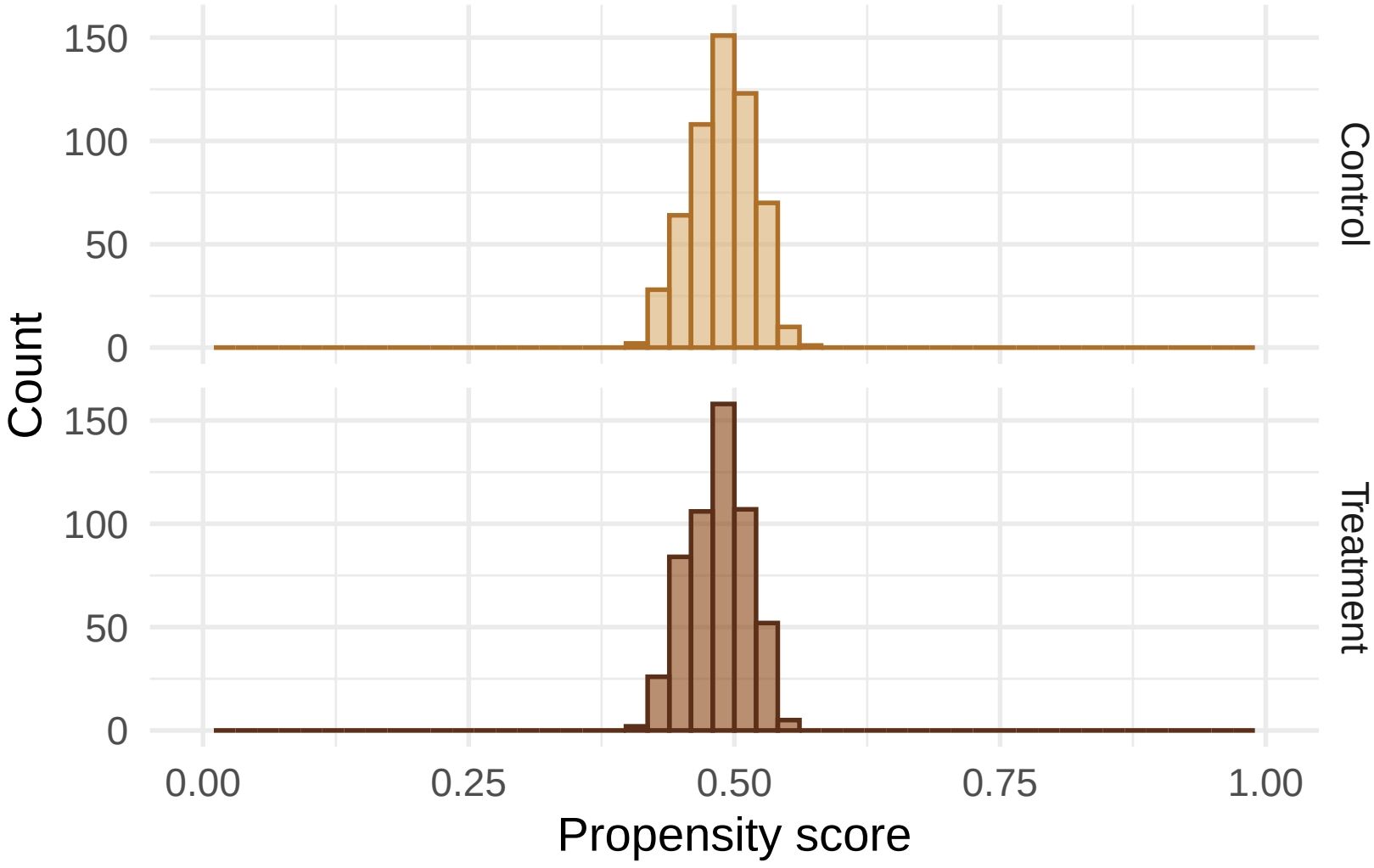
## Propensity model
W.hat.mod <- regression_forest(X = predictors,
                               Y = W,
                               clusters = clusters,
                               num.trees = 2000,
                               tune.parameters = "all",
                               num.threads = 2)

W.hat.rf <- W.hat.mod$predictions

## Outcome model
Y.hat.mod <- regression_forest(X = predictors,
                               Y = Y_cldmc,
                               clusters = clusters,
                               num.trees = 2000,
                               tune.parameters = "all",
                               num.threads = 2)

Y.hat.rf <- Y.hat.mod$predictions
```

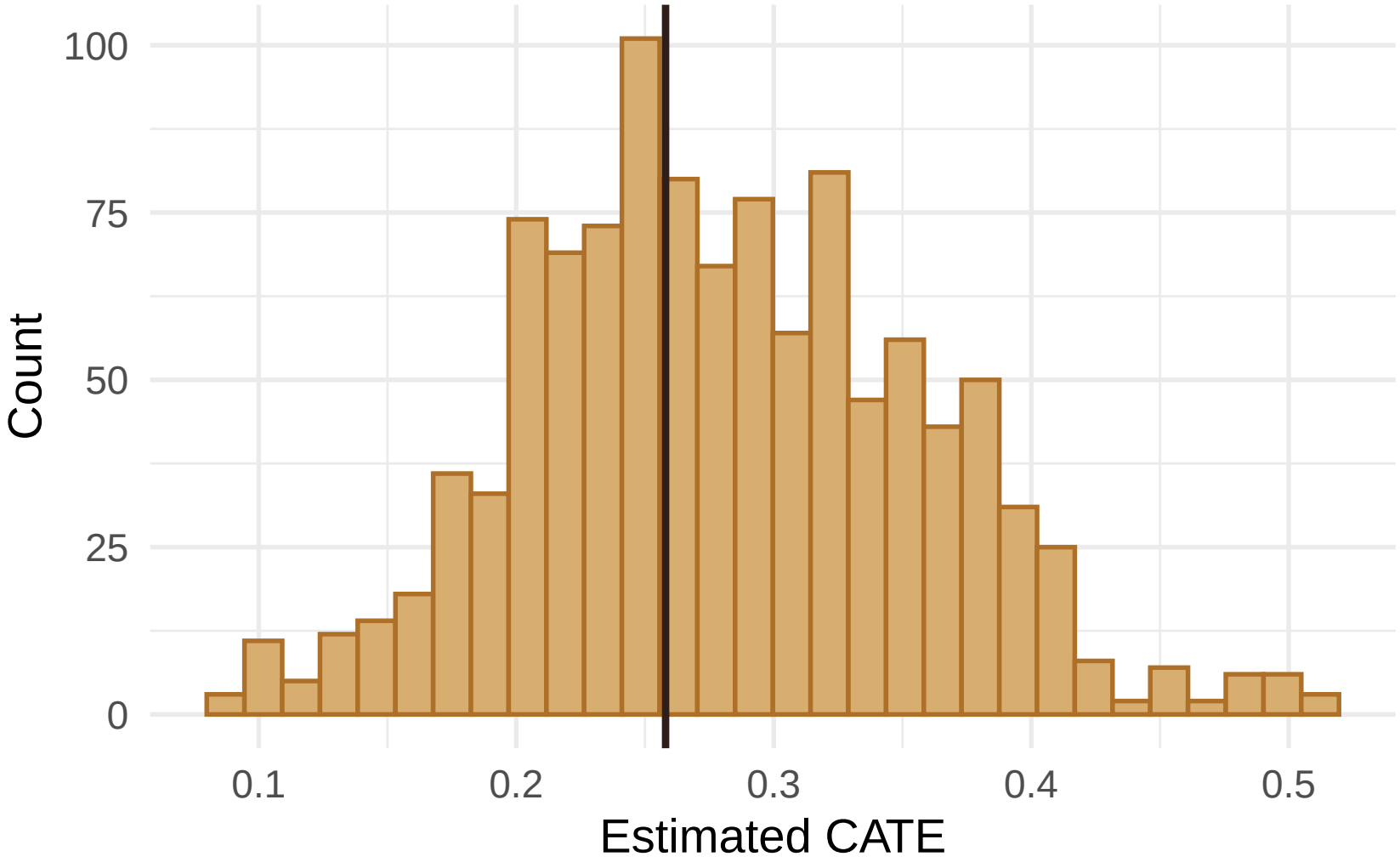
Verifying Randomization/Balance/Overlap



Estimating the Causal Forest

```
cf <- causal_forest(X = predictors,
                    Y = Y_cldmc,
                    W = W,
                    Y.hat = Y.hat.rf,
                    W.hat = W.hat.rf,
                    clusters = clusters,
                    num.trees = 2000,
                    tune.parameters = "all",
                    num.threads = 2)
```

Distribution of Estimated CATEs



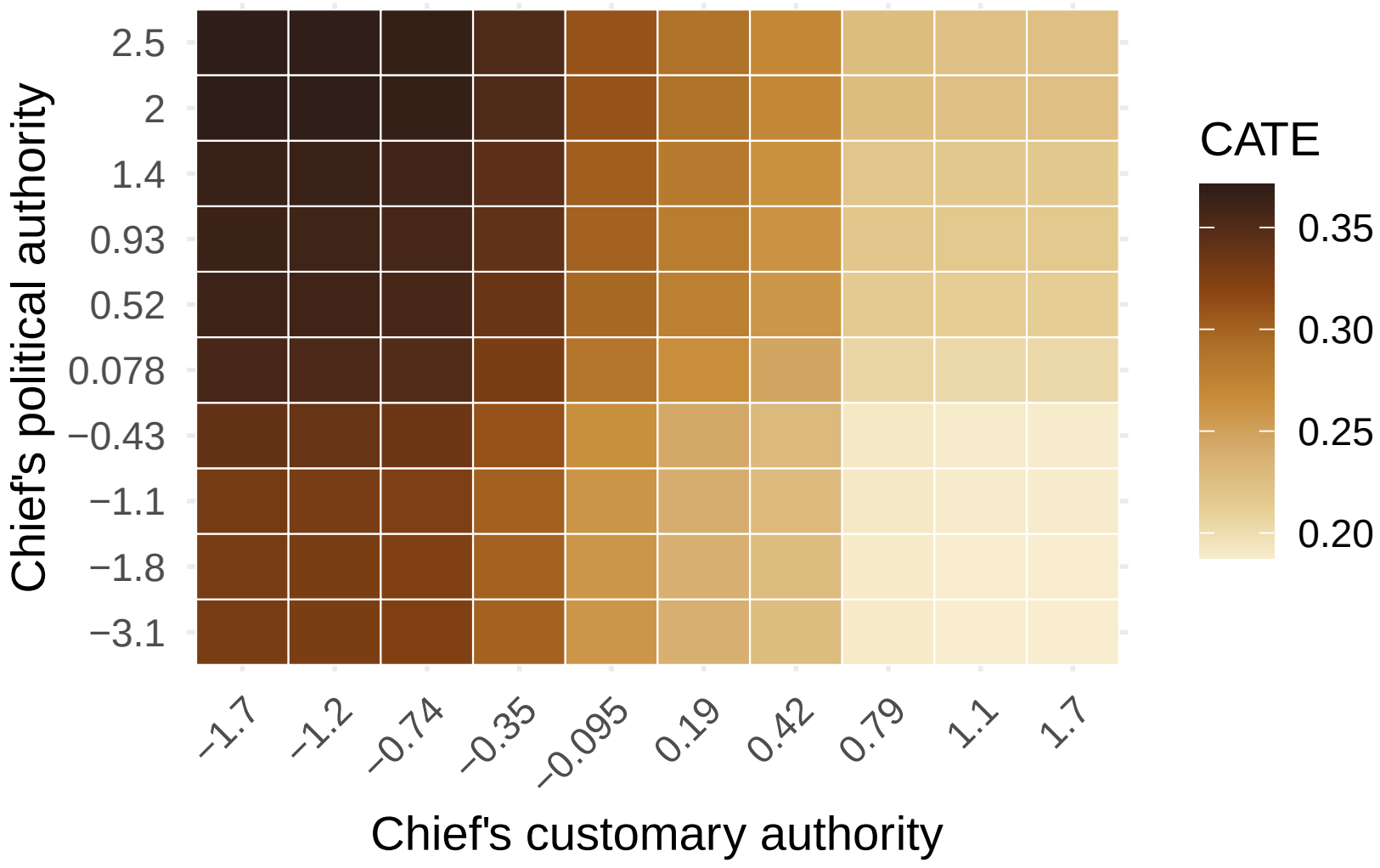
This process also tells us what variables are most important

```
importance <- variable_importance(cf)
names(importance) <- colnames(predictors)
data.frame(importance, colnames(predictors))

  importance   colnames.predictors.
1 0.09207416     chief_relationship
2 0.25791576 participate_index_pca1
3 0.28645619 participate_index_pca2
4 0.23300417      centroid_distance
5 0.01897067             cldmc_exist
6 0.11157904              cldmc_know
>
```

- We can scores each covariate by how often the forest splits on it.
- Early splits count more: depth k is weighted by k^{-2}, down to depth 4. Scores sum to 1.
- Descriptive, not a test: correlated covariates split the credit.

CATEs Across the Two Most Important Covariates



Testing Calibration

```
test_calib_orthog <- test_calibration(cf)
test_calib_orthog

Best linear fit using forest predictions (on held-out data)
as well as the mean forest prediction as regressors, along
with one-sided heteroskedasticity-robust (HC3) SEs:

                               Estimate Std. Error t value  Pr(>t)
mean.forest.prediction          0.92103    0.40912  2.2512 0.01228 *
differential.forest.prediction  0.16572    1.47781  0.1121 0.45537
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1
```

The Pseudo-Outcome ρ_i

ρ_i is observation i's contribution to $\hat{\tau}_P$, so for a candidate child C:

$$\hat{\tau}_C \approx \hat{\tau}_P + \big(\text{average of } \rho_i \text{ in } C\big)$$

Cutting where average ρ_i differs most is close to cutting where treatment effects differ most, without re-estimating τ in every candidate child. For each observation i in the node P being cut:

$$\rho_i = \frac{1}{A_P} \underbrace{(W_i - \bar{W}_P)}_{\text{(a)}} \Big[\underbrace{(Y_i - \bar{Y}_P)}_{\text{(b)}} - \underbrace{(W_i - \bar{W}_P)\,\hat{\tau}_P}_{\text{(c)}} \Big]$$

(a)	Recentered treatment
(b)	Recentered outcome
(c)	What the node's treatment effect $\hat{\tau}_P$ predicts for observation i
(b)−(c)	The residual from that prediction
A_P	Variance of treatment in the node; only rescales

$\rho_i > 0$ for a treated observation that did better than predicted, or a control that did worse. Both point to an effect larger than $\hat{\tau}_P$. Back

Estimating the Variance of $\hat{\tau}(x)$

$\hat{\tau}(x)$ is the value that sets the weighted score $\Psi = \sum_i \alpha_i(x)\,\psi_i$ to zero, so its variance is the noise in Ψ divided by how strongly Ψ responds to τ:

$$\hat{\sigma}^2(x) \;\approx\; \frac{\hat{H}(x)}{\hat{V}(x)^2}$$

$\hat{H}$ comes from growing the trees in bags of ℓ, where each bag uses its own random half of the data:

$$\hat{H}(x) \;=\; \underbrace{\mathrm{Var}_g\left(\bar{\Psi}_g\right)}_{\text{(a)}} \;-\; \frac{1}{\ell - 1}\,\underbrace{\mathrm{avg}_g\,\mathrm{Var}_{b \in g}\left(\Psi_b\right)}_{\text{(b)}}$$

ψ_i	Observation i's residual times its recentered treatment
$\Psi_b,\ \bar{\Psi}_g$	Ψ from tree b alone; its average over the trees in bag g
(a)	Variance between bags: sampling noise plus tree-growing noise
(b)	Variance within bags: tree-growing noise only
$\hat{V}$	Variance of treatment among the neighbors of x

Subtracting leaves the variation from which half of the data a bag saw: the sampling variance. A 95% interval is $\hat{\tau}(x) \pm 1.96\,\hat{\sigma}(x)$. Back